

ALLIANCE, NE, USA

WMO: 725635

Lat: 42.057N

Lon: 102.802W

Elev: 3929

StdP: 12.73

Time Zone: -7.00 (NAM)

Period: 96-19

WBAN: 24044

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-8.2	-1.6	-13.9	3.0	-7.5	-7.6	4.2	-0.9	34.0	30.6	31.2	30.5	6.4	280	0.697

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	30.6	97.0	65.1	93.2	64.9	90.3	64.5	70.0	85.5	68.6	84.0	67.3	82.7	12.3	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.8	110.4	74.4	64.1	103.7	73.2	62.9	99.6	72.5	36.5	85.2	35.1	83.7	33.9	82.1	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 30.6	26.3	23.6	DB	-20.4	101.9	6.0	3.2	-24.7	104.1	-28.3	106.0	-31.6	107.8	-36.0	110.1	(4)
(5)			WB	-21.0	73.0	5.9	2.1	-25.2	74.5	-28.6	75.7	-31.9	76.9	-36.2	78.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.2	25.1	27.1	36.9	44.6	54.1	66.0	73.5	70.7	61.2	46.4	34.6	24.6
	DBStd	19.55	11.41	11.46	10.65	9.29	8.64	7.33	5.51	5.92	8.49	9.69	10.53	11.68
	HDD50	3543	771	641	417	208	59	1	0	0	12	179	466	789
	HDD65	7144	1236	1061	871	613	347	75	8	20	169	578	912	1254
	CDD50	2512	0	0	11	45	187	480	727	643	348	67	4	0
	CDD65	639	0	0	0	0	10	104	270	198	56	1	0	0
	CDH74	9791	0	0	17	71	364	1636	3649	2726	1172	150	6	0
	CDH80	4647	0	0	0	8	108	714	1939	1341	501	36	0	0
	WSAvg	10.8	10.7	11.2	11.9	12.6	11.7	10.8	9.6	9.6	9.9	10.6	10.3	10.5
	PrecAvg	15.5	0.2	0.3	0.6	1.8	2.8	2.9	2.3	1.6	1.3	0.9	0.4	0.2
(15) Precipitation	PrecMax	21.7	0.7	0.7	1.6	3.4	9.4	6.6	4.7	2.9	3.0	2.8	0.9	0.9
	PrecMin	7.2	0.0	0.0	0.0	0.2	0.9	0.5	0.7	0.3	0.2	0.0	0.0	0.0
	PrecStd	3.1	0.2	0.2	0.5	0.9	1.9	1.6	0.9	0.7	0.8	0.6	0.2	0.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.7	63.8	76.6	80.8	88.4	98.8	101.8	99.2	95.2	84.4	73.5	61.5
		MCWB	43.4	44.9	50.5	54.4	60.5	64.5	65.8	64.9	63.4	56.7	49.8	43.5
	2%	DB	54.3	56.8	69.8	75.3	82.4	92.7	98.0	95.0	90.5	78.6	66.5	54.2
		MCWB	40.2	41.9	48.4	52.6	59.7	64.4	65.7	64.2	61.5	54.5	46.9	40.7
	5%	DB	47.7	50.5	63.7	70.3	78.0	88.3	94.4	91.3	86.2	72.3	60.8	47.7
		MCWB	37.3	38.8	45.8	51.1	58.3	64.1	65.9	64.4	60.6	52.4	44.5	37.1
	10%	DB	42.5	45.2	57.2	63.9	72.8	83.3	90.7	88.2	81.3	66.0	53.9	42.6
		MCWB	34.6	35.8	43.1	48.5	56.4	63.4	66.0	64.5	59.5	50.3	41.6	34.3
	0.4%	WB	44.4	46.3	52.9	57.1	65.4	71.3	72.2	71.1	67.4	59.5	51.1	44.9
		MCDB	58.0	62.3	70.4	73.9	79.1	84.2	87.5	87.0	84.4	76.5	70.1	60.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	40.9	42.4	49.4	54.0	62.5	68.6	70.5	69.0	65.0	56.5	47.6	41.1
		MCDB	53.1	55.5	66.6	70.8	75.8	82.2	86.5	84.5	81.5	73.4	65.0	53.1
	5%	WB	37.7	39.3	46.6	51.8	60.2	66.7	69.2	67.6	63.3	53.7	45.0	37.6
		MCDB	46.6	48.9	62.1	67.7	73.5	81.3	85.1	82.9	78.8	68.4	59.7	47.1
	10%	WB	35.0	36.3	43.9	49.6	57.8	64.9	68.0	66.2	61.5	51.3	41.8	34.6
		MCDB	41.7	44.4	56.8	62.9	70.4	79.7	83.8	81.3	76.6	64.5	52.9	41.8
(35) Mean Daily Temperature Range	5% DB	MDBR	26.5	26.1	28.9	28.1	27.8	29.2	30.6	30.6	31.8	30.5	29.9	26.9
		MCDBR	33.8	34.6	40.8	40.4	37.1	36.4	37.4	36.4	39.7	42.4	41.1	34.2
		MCWBR	21.4	21.1	22.0	20.2	16.9	13.3	11.6	11.9	15.3	21.3	23.6	22.0
	5% WB	MCDBR	31.3	31.9	38.3	36.7	31.3	30.9	30.3	30.4	33.0	36.5	39.3	32.7
		MCWBR	20.5	20.2	21.4	19.1	15.6	14.1	11.5	11.5	13.8	19.2	22.9	21.6
(40) Clear-Sky Solar Irradiance	taub		0.249	0.256	0.280	0.307	0.332	0.341	0.369	0.361	0.322	0.283	0.260	0.250
	taud		2.570	2.549	2.512	2.440	2.443	2.443	2.403	2.405	2.495	2.577	2.558	2.544
	Ebn at Noon		295	308	310	306	298	295	285	284	290	293	287	284
	Edh at Noon		21	25	30	34	35	36	36	35	30	24	21	20
(44) All-Sky Solar Radiation	RadAvg		706	985	1365	1688	1974	2313	2286	1983	1601	1100	773	594
	RadStd		37	54	101	99	155	141	80	71	83	97	44	36

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page